6. Baseband data communication systems

1. Introduction to information theory- measure of information, source entropy, symbol rate and data (information) rates
2. Shannon Hartley channel capacity theorem, implications and limitations
3. Line codes- types, properties and comparison
4. Baseband data communication structure, inter-symbol interference (ISI)
5. Pulse shaping techniques for zero ISI – raised cosine, duo-binary (correlative) encoding techniques
6. M-ary signaling, comparison with binary signaling
7. Eye Diagram

Introduction to information theory:

- Information is the source of a communication system, whether it is analog or digital. It is the intelligence/idea or message in information theory. Message may be in the form of electrical signal, speech or voice, picture or image, video and text.
- Information theory is a mathematical approach to the study of coding of information along with the quantification, storage, and communication of information.
- Information theory is also responsible for measuring and representing information, as well as the processing capacity of communication systems to transmit that information.
- The mathematical theory of information was proposed in 1949 by mathematician and engineer Claude Shannon and biologist Warren Weaver. However, it was the result of research initiated almost thirty years prior by scientists such as Andrei Markovi and Ralph Hartley; the latter is known for being one of the first representatives of binary language.
- Information theory deals with both theoretical and practical aspects of data compression and reliable transmission of information over noisy channels. The data source entropy gives a lower bound for the rate of data compression. Rates for reliable information transmission are bounded by the capacity of the given channel. The theory also includes theoretical analysis of secrecy systems.

Elements of information theory:

1. **Source of information or sender:** the party capable of issuing a message. In information theory, the main sources are:
2. **Random:** when the message cannot be predicted.
3. **Structured:** when there is a certain level of redundancy and order.
4. **Unstructured:** when all messages are random, unrelated, and meaningless, so there is a loss of part of the message.
5. **Message:** a set of data transported through a channel.
6. **Code:** a set of elements that follow a set of rules to be combined so they can be interpreted.

7. **Channel:** the medium by which the message is transmitted so that it reaches the receiver.
8. **Information:** what the sender seeks to convey through a message. From the point of view of mathematical probability, a theoretical framework of information theory, information should be proportional to the number of *bits* needed to recognize the message.
9. **Recipient:** the party that receives the message. Being able to assimilate the content of the message that originates from the source or sender is essential.
10. **Noise:** different causes that prevent the message from arriving normally in the information flow process, impeding the receiver from being able to understand it fully.

General approaches of information theory:

- It allows us to study the information process, such as the possible communication channels and understanding the data sent.
- It determines the simplest, most effective way to send a message without any alterations in the process.
- It recognizes elements of distortion or impediments in order for a message to reach a recipient optimally.
- It establishes that both sender and receiver should be able to encode and decode the messages.
- It analyzes the speed with which the messages are transmitted.
- It includes the possibility that a message has multiple meanings, so the recipient gives it meaning, as long as they have the same code as the sender.
- If the choice of message is presented as one of only two options, the value of the information is equal to one (a unit called a *bit*). In order for the information to be a *bit*, there must be equally likely alternatives in the process. In addition, it states that there is more information with the more alternatives there are, and all are equally likely.

Applications of information

- Information theory represents one of the most important branches of applied mathematics. Some of its various uses include:
- Computer science, such as cryptography and understanding data.
- Electrical engineering, such as communication theory and coding theory.
- Statistics.
- Biology, in the study of DNA sequences and the genetic code.
- Payments, electronic transactions, and authentication processes.
- Encrypting messages or Steganography (Steganography is the technique of hiding data within an ordinary, non secret file or message to avoid detection; the hidden data is then extracted at its destination).

Measure of information: The information source can be classified as having memory or memory less. A source with memory is one for which the occurrence of current symbol depends on previous symbols. A memory less source is one for which each symbol produced is independent of the previous symbols. Let us assume a discrete memory less source with 'q' possible message represented by { x_1, x_2, ………x_n} with probabilities of occurrence { p_1, p_2, ………p_n} such that $p_1 + p_2 + \ldots\ldots + p_n = 1$.

- Let I(x) be the amount of information content in the i^{th} message. Based on intuition, for $I(x_i)$ to represent information content of x_i message, the following condition of equation should be met.
- $I(x_i) > I(x_j)$, if $p_i < p_j$ means that the information content of event with less probability is higher than that of with high probability occurrence.
- In equation $I(x_i) \rightarrow 0 \; if \; p_i \rightarrow 1$ means that the information content in an event with less probability of occurrence near to unity (also called 'almost certain' event) is near to zero. The example is the message to a person living in earth that "Sun rises from east".
- In equation $I(x_i) \rightarrow 1 \; if \; p_i \rightarrow 0$ means that the information content in the message about an event which is 'almost impossible' "Sun rises from west".
- In equation $I(x_i) \geq 0 \; when \; 0 \leq p_i \leq 1$ means that message should contain some information i.e $I(x_i)$ should be non negative (i.e there should not be any information in a message that would increase uncertainty in the recipient after receiving messages)
- In equation $I(x_i \text{ and } x_j) = I(x_i \, x_j) = I(x_i) + I(x_j)$ means if $_i$ and x_j are two statistically independent messages coming from the same source then total information received from two message is equal to the sum of information contents in each message.
- The following relationship between $I(x_i)$ and p_i would satisfy all the above conditions for $I(x_i)$ to represent the information contained by message x_i with the probability of occurrence p_i.

  $$I(x_i) = log_2(\frac{1}{p_i})$$

  $$and \; log2 \; x = \frac{log_{10} x}{log_{10} 2}$$

- If two bit (binary digits) 1 and 0 occur with equal probabiliy and are correctly detected at the reciving end,then the information content in each digit is 1 bit.

  $$I(o \text{ or } 1) = - log_2(\frac{1}{2}) = 1 \text{ bit}$$

Properties of information:

- More uncertainty about message $\longrightarrow$ Information is more
- Receiver knows message being transmitted $\longrightarrow$ Information is zero
- I_1 is the information of message m_1 and I_2 is of m_2 $\longrightarrow$ $(I_1 + I_2)$ combined information by m1 and m2.
- If there are $M = 2^N$ equally likely messages then amount of information carried by each message will be $\longrightarrow$ N bits.

Proof for property 1: Suppose we have two different symbols m_1 and m_2 and their probabilities of happening are ¼ and ¾ (uncertainty if m_1 is greater than uncertainty of m_2 i.e. $U_{m1} > U_{m2}$) if we are able to prove that $I_{m1} > I_{m2}$ then property 1 is true.

We have $I(m_i) = log_2(\frac{1}{p_i})$

or $I(m_1) = log_2(\frac{1}{\frac{1}{4}})$

or $I(m_1) = 2$

Similarly

$I(m_2) = log_2(\frac{1}{\frac{3}{4}})$

or $I(m_2) = 0.415$

Hence proved

Proof for property 2: If the receiver knows the message being transmitted then the information is zero.

$P_k = 1$

then $I_k = log_2(\frac{1}{p_k})$

or $I_k = log_2(\frac{1}{1})$

$I_k = log_2(1) = 0$ bits.

Hence it is proved if the receiver knows the message being transmitted then the information is zero.

- **Proof for property 3:** If I_1 is the information of message m_1 and I_2 is of m_2 then $(I_1 + I_2)$ is the combined information by m1 and m2.

  Let the probability of occurring I_1 message is p_1 and I_2 message is p_2 then

  $I_1 = log_2(\frac{1}{p_1})$ and $I_2 = log_2(\frac{1}{2})$

  Since the messages m_1 and m_2 are independent the probability of composite messages is p_1p_2.

  $I_{1,2} = log_2[\frac{1}{p_1p_2}]$

  $= log_2[(\frac{1}{p_1})(\frac{1}{p_2})]$

  $= log_2[\left(\frac{1}{p_1}\right) + log_2(\frac{1}{p_2})]$

  $= I_1 + I_2$ hence proved

Proof for property 3: If there are $M = 2^N$ equally likely messages then amount of information carried by each message will be N bits

Since we have 'M' equal likely messages then probability of each message will be $\frac{1}{M}$

$I_k = log_2(\frac{1}{p_k}) = log_2(\frac{1}{\frac{1}{M}}) = log_2(M)$ and we have $M = 2^N$

Hence $I_k = log_2(2^N) = N log_2(2) = N$ bits

Source Entropy

Source entropy measures the **average uncertainty** in an information source. It quantifies how much information is produced on average per symbol by a random source. Higher **entropy** means the source is more unpredictable, while **lower entropy** means it is more predictable.

Let the source emit one of k possible symbols (long independent sequence of symbols i.e. the length of symbols N in the sequence of much more greater than the possible symbols 'k') x_1 , x_2,….. x_k is statistically independent sequence i.e probability of occurrence of x_i does not depend upon previous occurrence of x_h or future occur of x_j with probabilities p_1, p_2……. P_k respectively. Now, the information content of individual symbol is discrete and random in nature (i.e cannot be determined), that takes on the value $I(x_1)$, $I(x_2)$……. $I(x_k)$. The mean value of information content $I(x_k)$ is called entropy and is denoted by H(X) where X is the discrete random variable.

$H(X) = E[I(x_k)]$

$H(X) = \sum_{k=1}^{k} p_k \, I(X_k)$

$H(X) = \sum_{k=1}^{k} p_k \, log_2(\frac{1}{p_k})$

Entropy is the measure of the average information content per source symbol. If the information is in bits then the unit of entropy id bits/symbol.

Symbol rate and data (information) rates

In digital communication and information theory, **symbol rate** and **data (information) rate** determine how efficiently data is transmitted over a channel.

Symbol Rate (Baud Rate)

The **symbol rate** (also called **baud rate**) is the number of symbols transmitted per second.

Formula:

$$R_S = \frac{R_b}{N}$$

where:

- Rs= Symbol rate (in baud or symbols per second),
- Rb= Bit rate (in bits per second),
- N = Number of bits per symbol (depends on modulation scheme).

Example:

- A **binary (2-level) system** (e.g., **NRZ encoding**: 0 or 1 per symbol):
  Rs = $\frac{1000\ bps}{1} = 1000$ baud
- A **4-level system** (e.g., **QPSK modulation**: 2 bits per symbol):
  Rs = $\frac{1000\ bps}{2} = 500$ baud

➢ **Higher modulation levels reduce symbol rate for the same data rate.**

Data Rate (Bit Rate or Information Rate)

The **data rate** (bit rate) measures the amount of digital information transmitted per second.

Formula:

$R_b = R_s \times N$

where:

- R_b = Bit rate (bps),
- R_s = Symbol rate (baud),
- N = Number of bits per symbol.

Example:

A system using **QPSK modulation** (2 bits per symbol) with a **symbol rate of 1000 baud**:

Rb=1000×2=2000 bps

So, the **bit rate is 2000 bits per second**.

➢ **Relationship Between Symbol Rate and Bit Rate**

- **Higher symbol rates** require **more bandwidth** but allow faster transmission.
- **Higher bit rates** can be achieved with **higher-order modulation** without increasing the symbol rate.
- **Efficient modulation** reduces bandwidth requirements while maintaining high bit rates.

Shannon Hartley channel capacity theorem:

The **Shannon-Hartley theorem** defines the **maximum channel capacity** (theoretical highest data rate) for a communication channel in the presence of noise. It is a fundamental principle in information theory, developed by **Claude Shannon** and **Ralph Hartley**.

Shannon-Hartley Theorem Formula:

- The channel capacity C (in bits per second, bps) is given by:

$$C = B \log_2\left(1+\frac{S}{N}\right)$$

Where:

- C = Channel capacity (maximum error-free data rate)
- B = Bandwidth of the channel (in Hz)
- S = Signal power (in watts)
- N = Noise power (in watts)
- $\frac{S}{N}$ = **Signal-to-Noise Ratio (SNR)** (can also be expressed in decibels, dB)
- Increasing bandwidth allows higher capacity, but not linearly (due to the logarithmic relationship).
- Represents the absolute maximum rate for reliable communication over a noisy channel.

Example Calculation:

If a channel has:

- Bandwidth B=1 MHz,
- SNR = 30 dB,

First, convert SNR from dB to linear scale:

$$\frac{S}{N} = 10^{\left(\frac{30}{10}\right)} = 1000$$

Now, compute capacity:

$$C = 10^6 \times log_2\,(1+1000) \approx 10^6 \times 9.967 \approx 9.967 \text{ Mbps}$$

Implications of the Shannon-Hartley Theorem

The Shannon-Hartley theorem has profound implications for communication system design, networking, and information theory. Key implications include:

1. Fundamental Limit on Data Rates

- Establishes the **maximum error-free transmission rate** for a given bandwidth and SNR.
- No coding or modulation scheme can exceed this limit without increasing bandwidth or SNR.

2. Adjustment between Bandwidth and Power

- **Increasing bandwidth** improves capacity, but only logarithmically with SNR.
- **Increasing signal power (SNR)** helps, but real-world systems face power constraints.
- Explains why **wideband systems (e.g., 5G, fiber optics)** achieve high speeds by using more bandwidth.

3. Role of Noise in Communication

- Noise fundamentally limits capacity, not just signal strength.
- **Low SNR → Low capacity**, even with infinite bandwidth (due to the logarithmic relationship).

4. Basis for Modern Communication Systems

- Guides the design of **error-correcting codes** (Turbo codes, LDPC) to approach Shannon's limit.
- Influences **modulation schemes** (QAM, OFDM) to maximize spectral efficiency.
- Used in **wireless (Wi-Fi, 5G), wired (DSL, fiber), and deep-space communications**.

5. Multi-User Implications (Extension to MIMO & Network Theory)

- In **MIMO (Multiple Input Multiple Output)**, capacity scales with the number of antennas (under certain conditions).
- Forms the basis for **network information theory**, studying capacity in multi-user channels (e.g., cellular networks).

Limitations of the Shannon-Hartley Theorem

While revolutionary, the theorem has key limitations in real-world scenarios:

1. Assumes Additive White Gaussian Noise (AWGN)

- Real channels suffer from **impulse noise, interference, and non-Gaussian distortions** (e.g., crosstalk in cables).
- **Fading (in wireless)** and multipath effects are not accounted for.

2. Ignores Channel Imperfections

- Does not consider **inter-symbol interference (ISI), phase distortion, or**
- **non-linearities** in amplifiers.
- Assumes a **static channel**, whereas real channels (e.g., mobile networks) are time-varying.

3. Assumes Infinite-Length Optimal Coding

- Achieving capacity requires **infinitely long, optimally designed codes**, which are impractical.
- Real-world codes (e.g., LDPC, Polar codes) **approach but never reach** the Shannon limit.

4. No Consideration of Latency or Complexity

- The theorem does not account for **encoding/decoding delay** or computational complexity.
- High-performance codes (near Shannon limit) often require **high latency and processing power**.

5. Power and Bandwidth Constraints in Practice

- **Regulatory limits** (e.g., FCC spectrum allocations) restrict bandwidth.
- **Energy efficiency** becomes critical in battery-powered devices (IoT, smart phones).

Line codes- types: A line code is the code used for data transmission of a digital signal over a transmission line. This process of coding is chosen so as to avoid overlap and distortion of signal such as inter-symbol interference.

Properties of Line Coding

- As the coding is done to make more bits transmit on a single signal, the bandwidth used is much reduced.
- For a given bandwidth, the power is efficiently used.
- The probability of error is much reduced.
- Error detection is done and the bipolar has a correction capability.
- Power density is much favorable.
- The timing content is adequate.
- A long string of 1s and 0s is avoided to maintain transparency.

Types of Line Coding

There are 3 types of Line Coding

1. Unipolar
2. Polar
3. Bi-polar

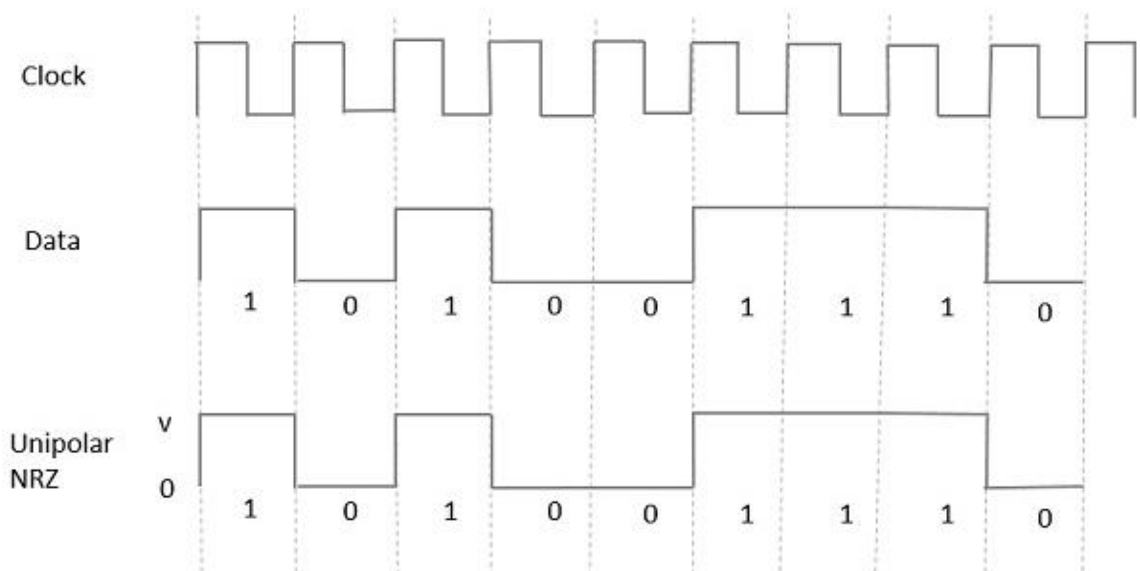
1. Unipolar line coding

- Unipolar line codingis also called as **On-Off Keying** or simply **OOK**.
- The presence of pulse represents a **1** and the absence of pulse represents a **0**.

There are two variations in Unipolar coding:

- **Non Return to Zero (NRZ)**

In this type of unipolar coding, a High in data is represented by a positive pulse called as Mark, which has a duration T_0 equal to the symbol bit duration. A Low in data input has no pulse.



Advantages

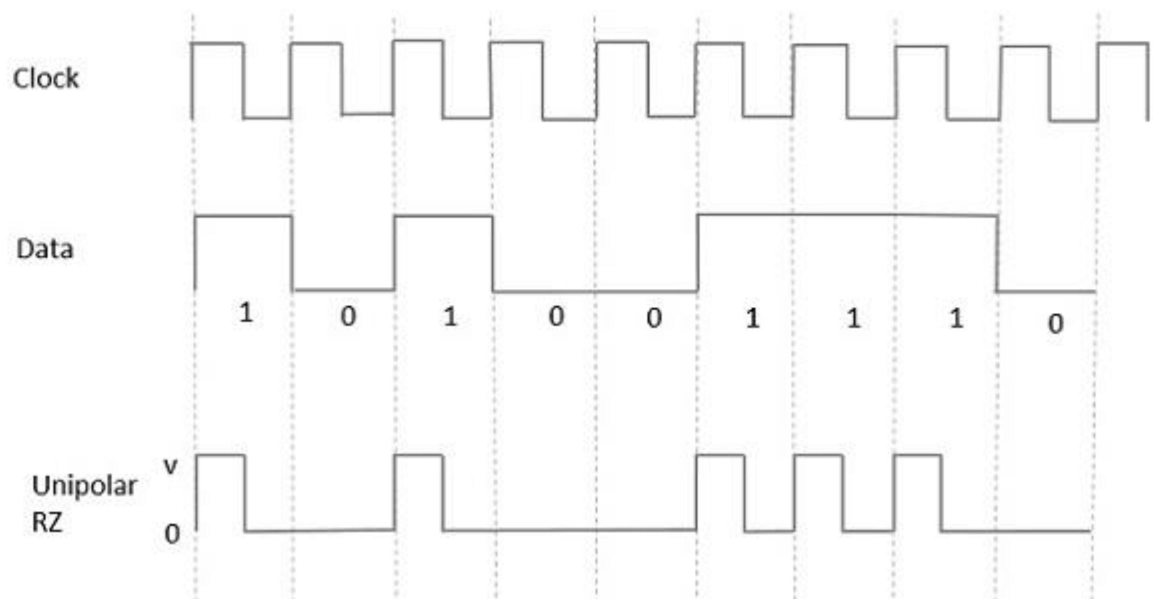
- It is simple.
- A lesser bandwidth is required.

Disadvantages

- No error correction done.
- Presence of low frequency components may cause the signal droop.
- No clock is present.
- Loss of synchronization is likely to occur (especially for long strings of 1s and 0s).

Return to Zero (RZ) :

In this type of unipolar coding, a High in data, though represented by a Mark pulse, its duration T_0 is less than the symbol bit duration. Half of the bit duration remains high but it immediately returns to zero and shows the absence of pulse during the remaining half of the bit duration.



Advantages

- It is simple.
- The spectral line present at the symbol rate can be used as a clock.

Disadvantages

- No error correction.
- Occupies twice the bandwidth as unipolar NRZ.
- The signal sink is caused at the places where signal is non-zero at 0 Hz

.(The **"signal sink"** (or **DC component problem**) occurs when a line-coded signal has **non-zero energy at 0 Hz (DC)**. This happens when the signal has **unequal numbers of positive and negative pulses** over time).

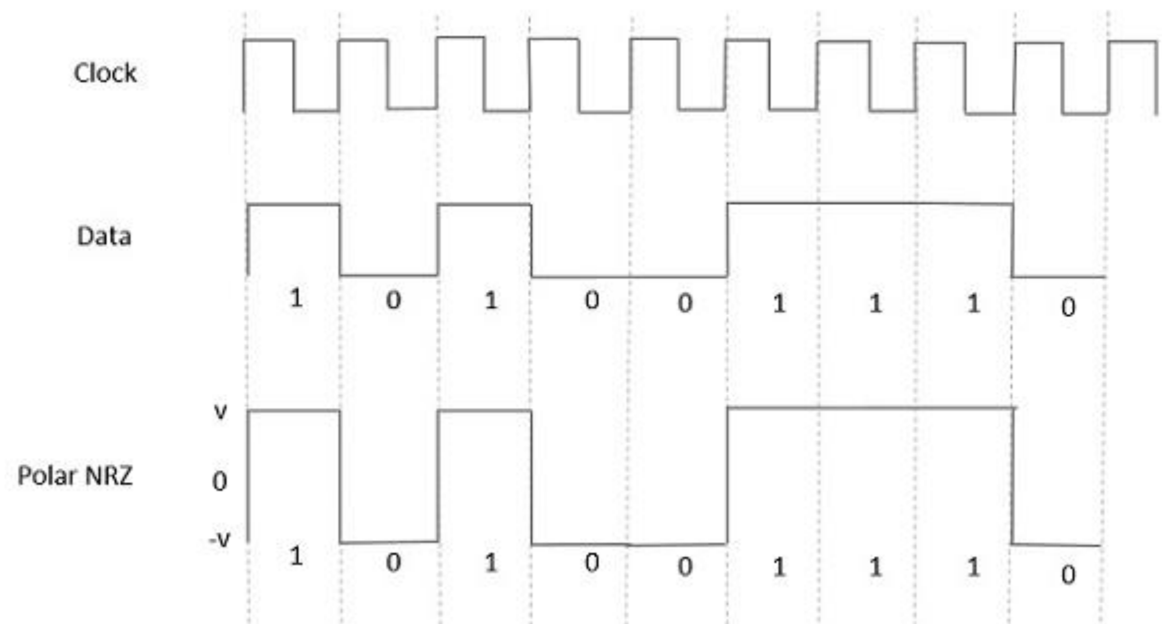
Polar Signaling

There are two methods of Polar coding. They are:

- **Polar NRZ**
- **Polar RZ**

➢ **Polar NRZ**

In this type of Polar signaling, a High in data is represented by a positive pulse, while a Low in data is represented by a negative pulse. The following figure depicts this well.



Advantages:

- It is simple.
- No low-frequency components are present.

Disadvantages:

- No error correction.
- No clock is present.
- The signal droop is caused at the places where the signal is non-zero at 0 Hz.

Bipolar Coding:

Bipolar coding is a **line coding** technique that uses **three voltage levels** (+V, 0, -V) to represent binary data. Unlike unipolar (one polarity) and polar (two polarities) coding, bipolar coding alternates pulses to **eliminate the DC component**, making it suitable for **AC-coupled systems** like telephone lines.

Features of Bipolar Coding

- **Three voltage levels**: +V (positive), 0 (zero), -V (negative).
- **No DC component**: Alternating pulses cancel out DC bias.
- **Built-in error detection**: Violation of alternation indicates an error.
- **Used in telecom systems**: T1/E1 lines, ISDN.

Types of Bipolar Coding

1. Alternate Mark Inversion (AMI)

- **1** → Alternates between **+V and -V** (marks).
- **0** → **0V** (space).
- **Example:**
-

Data	1	0	1	1	0	1	0	0	1
AMI	+V	0	-V	+V	0	-V	0	0	+V

Advantages:

- No DC component.
- Simple error detection (two consecutive 1s with same polarity = error).

Disadvantages:

- Long 0 sequences lose synchronization (fixed with **B8ZS/HDB3** scrambling).

Applications:

- T1/E1 carrier systems.

2. Pseudoternary (Inverse of AMI)

- **0** → Alternates between **+V and -V**.
- **1** → **0V**.
- **Example:**

Data	1	0	1	1	0	1	0	0	1
AMI	0	+V	0	0	-V	-0	+V	-V	0

Advantages:

- Same DC-free benefits as AMI.

Disadvantages:

- Long 1 sequences cause sync issues (rarely used alone).

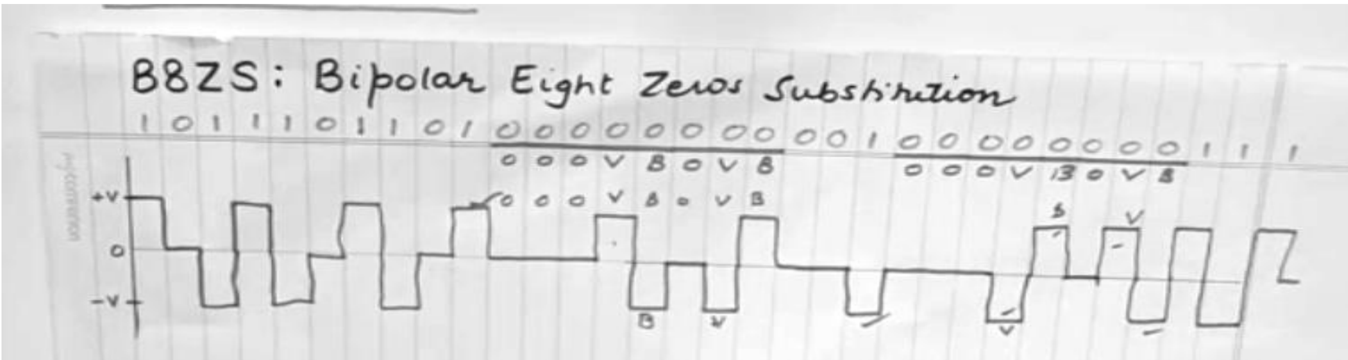
3. Bipolar with Scrambling (B8ZS / HDB3)

Since AMI struggles with long 0 sequences, **scrambling techniques** are used:

a) B8ZS (Bipolar with 8-Zero Substitution)

- Replaces **8 consecutive 0s** with a special pattern:
  - 000VB0VB (where V = violation pulse, B = bipolar pulse).
- **Example:**

Data	1	0	0	0	0	0	0	0	0	1
AMI	+V	0	0	0	+V	-V	0	-V	+V	0



If last pulse is negative '-'	0	0	0	-	+	0	+	-
If last pulse is positive '+'	0	0	0	+	-	0	-	+

b) HDB3 (High-Density Bipolar 3-Zero Substitution)

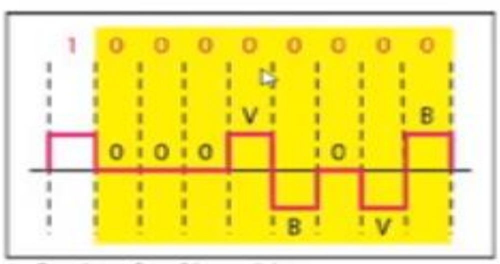
- Replaces **4 consecutive 0s** with a pattern ensuring no DC bias:
  - If odd # of 1s since last violation → 000V.
  - If even # of 1s → B00V.

	Odd	**Even**
Last pulse negative '-'	000-	+00+
Last pulse positive '+'	000+	-00-

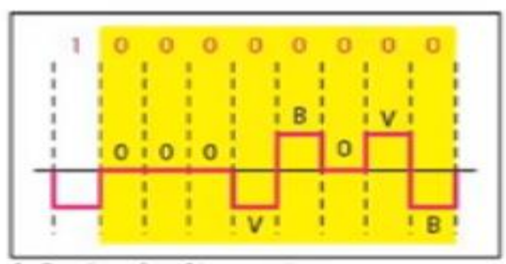
- **Example:**

Data	1	0	0	0	0	1	0	0	0	0	1
HDB3	+V	0	0	0	+V	-V	0	0	0	-V	+V

B8ZS

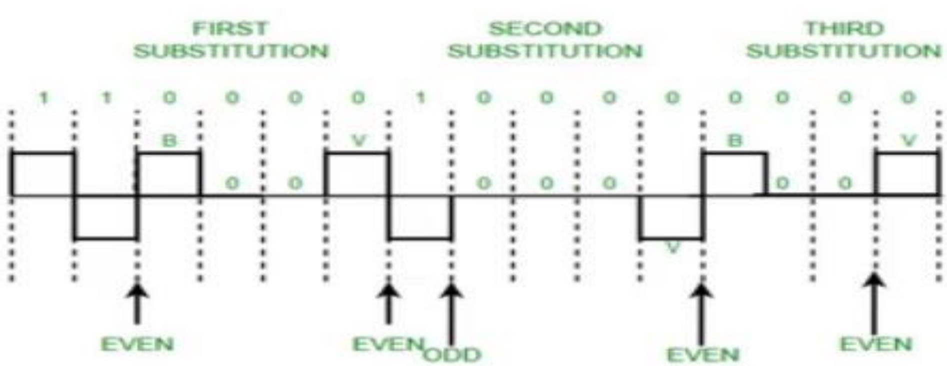


a. Previous level is positive.

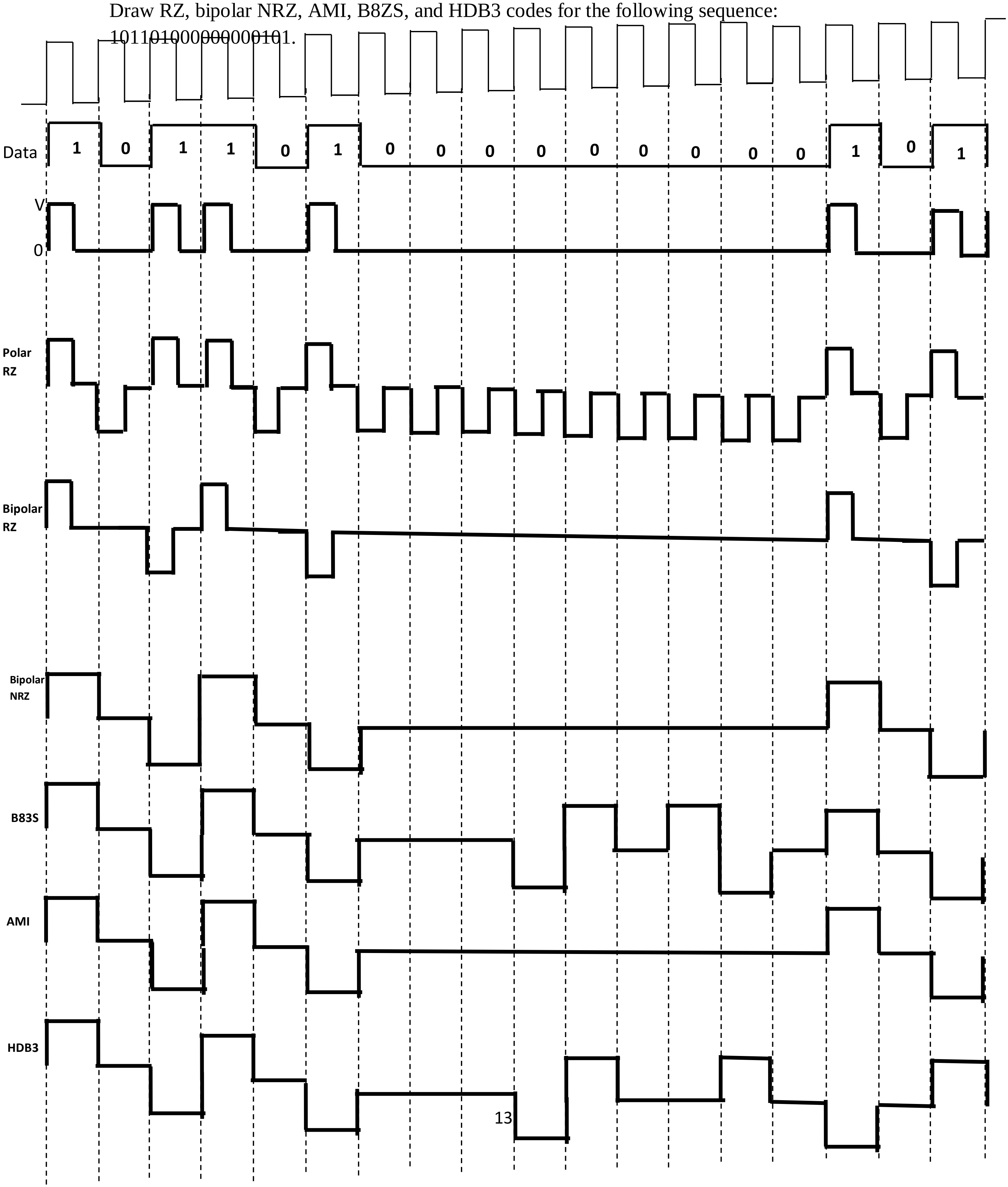


b. Previous level is negative.

HDB3



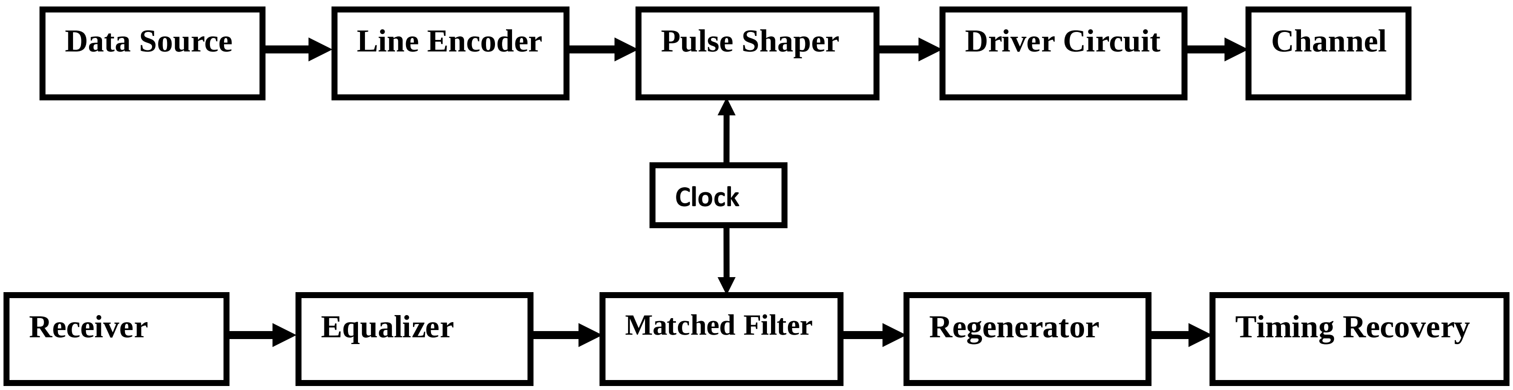
Draw RZ, bipolar NRZ, AMI, B8ZS, and HDB3 codes for the following sequence: 101101000000000101.



Baseband Data Communication System Structure

- Baseband communication transmits **digital signals directly over a channel** without modulation (i.e., no carrier signal). It is widely used in **wired networks (Ethernet, USB), short-distance links, and digital interfaces**.
- In baseband data communication, information is transmitted over a channel using signals that occupy the entire bandwidth of the channel without modulation to shift them to a different frequency range.

Block Diagram



Transmitter Section

a) Data Source

- Generates raw binary data (e.g., from a computer, sensor, or storage device).

b) Line Encoder

➢ Converts bits into **voltage waveforms** using schemes like:

- **NRZ, RZ** (simple but DC issues)
- **Manchester** (self-clocking, no DC)
- **AMI/B8ZS** (telecom, DC-free)

c) Pulse Shaping Filter

- Shapes pulses to **minimize Inter-Symbol Interference (ISI)**.
- Common filters: **Raised Cosine, Nyquist**.
- Ensures signals fit within **channel bandwidth**.

d) Driver Circuit

- Amplifies signal for transmission.
- Matches **impedance** to the channel (critical for signal integrity).

2. Transmission Channel

- **Physical medium**: Twisted pair (Ethernet), coaxial cable, PCB trace, optical fiber.
- Introduces:
  - ❖ **Attenuation** (signal loss over distance)
  - ❖ **Noise** (thermal, crosstalk)
  - ❖ **Distortion** (ISI, phase shifts)

3. Receiver Section

a) Regenerator (Repeater/Amplifier)

- Boosts weakened signals (but also noise).

b) Timing Recovery

- Extracts **clock signal** from data (critical for sampling).
- Methods:
  - **PLL (Phase-Locked Loop)**
  - **Preamble sync** (e.g., Ethernet frames)
  - **Edge detection** (Manchester coding)

c) Matched Filter

- Optimizes **signal-to-noise ratio (SNR)**.
- Designed to match transmitted pulse shape.

d) Equalizer

- Compensates for **channel-induced distortion & ISI**.
- Types:
  - **Linear Equalizer** (simple FIR filter)
  - **Decision Feedback Equalizer (DFE)** (better for severe ISI)

e) Decision Circuit

- Samples signal at **optimal instants** (mid-bit).
- Compares to threshold → decides **0 or 1**.
- Errors occur if noise/ISI corrupts signal.

f) Data Sink

- Final destination (e.g., display, memory, network processor).

Comparison: Baseband vs. Broadband

Feature	**Baseband**	**Broadband**
Signal Type	Digital (raw bits)	Analog (modulated carrier)
Bandwidth Use	Entire channel	FDM/TDM (shared channels)
Modulation	Not used	Required (QAM, OFDM, PSK)
Distance	Short-medium (LANs, PCB traces)	Long (cable TV, cellular)
Examples	Ethernet, USB, HDMI	Wi-Fi, DSL, Cable Internet

Inter-Symbol Interference (ISI) in Digital Communication

- **Inter-Symbol Interference (ISI)** occurs when the **time dispersion (Time Dispersion** refers to the spreading or broadening of a signal over time as it propagates through a medium or a communication channel) of a transmitted signal causes adjacent symbols to overlap, corrupting data detection. It is a major challenge in **high-speed baseband communication** (e.g., Ethernet, DSL, optical fiber).
- Digital data is represented by electrical pulse, communication channel is always band limited. Such a channel disperses or spreads a pulse carrying digitized samples passing through it. When the channel bandwidth is greater than bandwidth of pulse, spreading of pulse is very less.
- But when channel bandwidth is close to signal bandwidth, i.e. if we transmit digital data which demands more bandwidth which exceeds channel bandwidth, spreading will occur and cause signal pulses to overlap. This overlapping is called **Inter Symbol Interference**.
- In short it is called ISI. Similar to interference caused by other sources, ISI causes degradations of signal. This problem of ISI exists strongly in Telephone channels like coaxial cables and optical fibers.

➢ The main objective is to study the effect of ISI, when digital data is transmitted through band limited channel and solution to overcome the degradation of waveform by properly shaping pulse

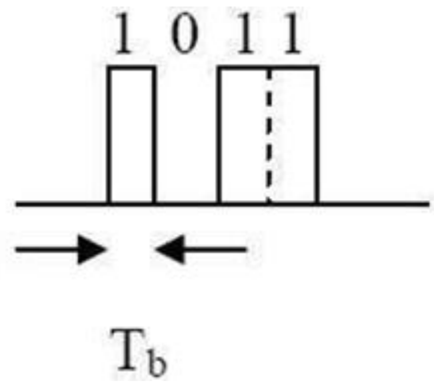


Fig: Transmitted waveform

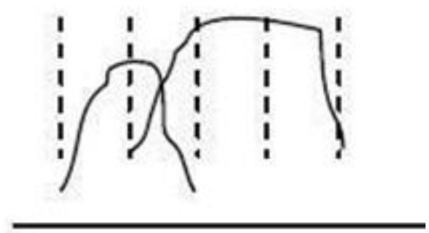
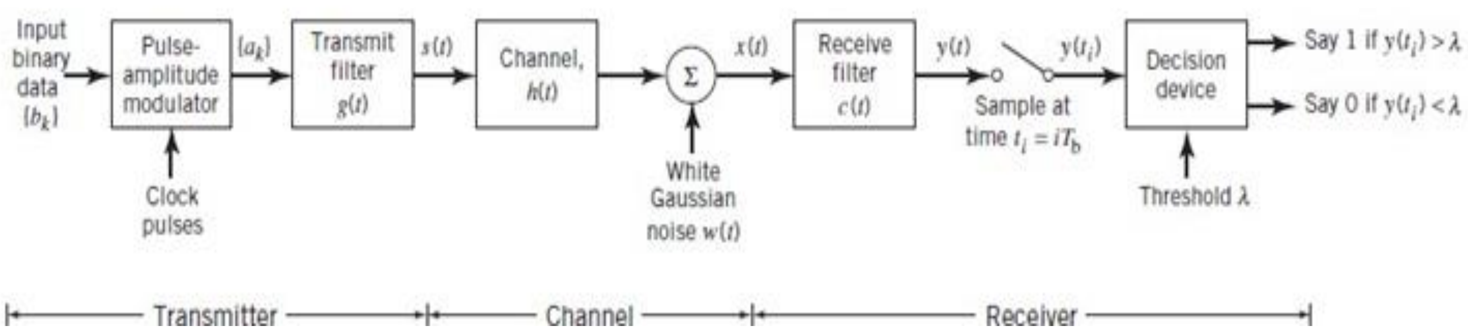


Fig: Pulse Dispersion

➢ The effect of sequence of pulses transmitted through channel is shown in fig. The Spreading of pulse is greater than symbol duration, as a result adjacent pulses interfere. i.e. pulses get completely smeared(damaged), tail of smeared pulse enter into adjacent symbol intervals making it difficult to decide actual transmitted pulse. First let us have look at different formats of transmitting digital data.



➢ In base band transmission best way is to map digits or symbols into pulse waveform. This waveform is generally termed as **Line codes**. To proceed with a mathematical study of inter symbol interference, consider a *baseband binary PAM system*, a generic form of which is depicted in Figure. The term "baseband" refers to an information- bearing signal whose spectrum extends from (or near)

➢ The *pulse-amplitude modulator* changes the input binary data stream {*bk*} into a new sequence of short pulses, short enough to approximate impulses. More specifically, the pulse amplitude *ak* is represented in the polar form:

$$a_k = \begin{cases} +1 \text{ if } b_k \text{ is symbol } 1 \\ -1 \text{ if } b_k \text{ is symbol } 0 \end{cases}$$

- The sequence of short pulses so produced is applied to a *transmit filter* whose impulse response is denoted by $g(t)$. The transmitted signal is thus defined by the sequence

$$s(t) = \sum_k a_k g(t - kT_b)$$

- A binary data stream represented by the sequence $\{a_k\}$, where $a_k = +1$ for symbol 1 and $a_k = -1$ for symbol 0, *modulates* the basis pulse $g(t)$ and superposes *linearly* to form the transmitted signal $s(t)$. The signal $s(t)$ is naturally modified as a result of transmission through the *channel* whose impulse response is denoted by $h(t)$. The noisy received signal $x(t)$ is passed through a *receive filter* of impulse response $c(t)$.
- The resulting filter output $y(t)$ is sampled *synchronously* with the transmitter, with the sampling instants being determined by a *Clock* or *timing signal* that is usually extracted from the receive-filter output. Finally, the sequence of samples thus obtained is used to reconstruct the original data sequence by means of a *decision device*. Specifically, the amplitude of each sample is compared with a *zero threshold*, assuming that the symbols 1 and 0 are equi-probable. If the zero threshold is exceeded, a decision is made in favor of symbol 1; otherwise a decision is made in favor of symbol 0. If the sample amplitude equals the zero threshold exactly, the receiver simply makes a random guess.
- Except for a trivial scaling factor, we may now express the receive filter output as

$$y(t) = \sum_k a_k p(t - kT_b)$$

where the pulse $p(t)$ is to be defined. To be precise, an arbitrary time delay t_0 should be included in the argument of the pulse $p(t - kT_b)$ in (8.6) to represent the effect of transmission delay through the system. To simplify the exposition, we have put this delay equal to zero without loss of generality; moreover, the channel noise is ignored.

Pulse shaping techniques for zero ISI – raised cosine:

- **Pulse shaping** is used in digital communication systems to **limit bandwidth** and **eliminate ISI** (Inter-Symbol Interference). ISI happens when the pulses from one symbol interfere with adjacent symbols, causing errors. To eliminate ISI, we use **pulse shaping techniques** that ensure pulses are zero at all other sampling instants except their own (**Nyquist criterion**).
- The Nyquist criterion provides a way to design pulse shapes that minimize or eliminate ISI. The Nyquist criterion states that the pulse shaping function, p(t), must satisfy the following condition:
  $p(iT_b - kT_b) = 1$ for $i = k$ and 0 for $i \neq k$, where T_b is the symbol period.

- **Raised Cosine Filter**

The **Raised Cosine (RC) filter** is designed to meet the Nyquist criterion. It smooths out the abrupt transitions of a rectangular pulse, which limits bandwidth and controls ISI.

Features:

- Zero ISI at sampling points
- Controlled bandwidth
- Tunable roll-off

Impulse Response:

$$h(t) = \frac{\sin(\pi t/T)}{\pi t/T} \cdot \frac{\cos(\pi \beta t/T)}{1-(2\beta t/T)^2}$$

Where:

- T: symbol period
- β: roll-off factor ($0 \leq \beta \leq 1$)
- β=0: Brick-wall filter (ideal sinc)
- β=1: Maximum roll-off
- β controls the trade-off between **bandwidth** and **time-domain smoothness**

Raised Cosine is Popular because:

- It **perfectly satisfies the Nyquist criterion**
- Smooth roll-off reduces spectral side lobes
- Practical to implement (e.g., root raised cosine used in transmit/receive filtering)

Duo-binary (correlative) encoding techniques:

- Duo-binary encoding is a **correlative coding** technique that introduces controlled **Inter-Symbol Interference (ISI)** to achieve higher spectral efficiency while maintaining reliable detection. Unlike Nyquist pulse shaping (e.g., raised cosine), which eliminates ISI, duo binary encoding **exploits ISI** in a predictable way to allow detection with simple methods.
- Instead of transmitting raw bits (say 1s and 0s), the transmitted signal **adds the current bit with the previous bit**. So each symbol **depends on the current and previous bit**.
- **Mathematical Representation**

The duo-binary encoder output y_k is given by: $y_k = x_k + x_{k-1}$

Where:

- x_k= current binary input (0 or 1)
- x_{k-1} = previous binary input
- y_k = encoded output (0, 1, or 2)

Example of Encoding

Input x_k	**Previous x_{k-1}**	**Output y_k**
0	0	0
0	1	1
1	0	1
1	1	2

Duo-binary Signaling & Detection
Frequency Response

- The duobinary scheme has a **cosine-shaped frequency response**:
  $H(f)=2T\cos(\pi f T), \quad |f| \le \frac{1}{2T}$
- **Bandwidth**: Occupies only **half the bandwidth** of Nyquist signaling ($B=\frac{1}{2T}$).

Detection Methods

Since yk has **three levels (0, 1, 2)**, detection can be done using:

1. **Threshold Detection**:
- If $yk=0 \rightarrow$ Decode as **0**
- If $yk=2 \rightarrow$ Decode as **1**
- If $yk=1 \rightarrow$ **Error or ambiguity** (requires differential decoding)

2. **Precoding (Avoid Ambiguity)**:
- Use **differential encoding** before duobinary encoding to remove ambiguity.
- Precoding rule: $dk=xk \oplus dk-1$
- Then, $yk=dk + dk-1$

Example with Precoding

Input xk	Precoded $d_k = x_k \oplus d_{k-1}$	$yk=d_k+d_{k-1}$	Decoded $x\hat{}k$
1	1	1 (initial)	-
0	$1 \oplus 0 = 1$	1 + 1 = 2	0 (since yk=2)
1	$1 \oplus 1 = 0$	0 + 1 = 1	1 (since yk=1)
0	$0 \oplus 0 = 0$	0 + 0 = 0	0 (since yk=0)

Advantages & Disadvantages

Advantages	Disadvantages
Higher spectral efficiency (half the bandwidth of Nyquist)	**Error propagation** (if one bit is wrong, subsequent bits may be affected)
Simple implementation (only addition/subtraction)	**Requires precoding** for reliable detection
No need for sharp cutoff filters (unlike raised cosine)	**Sensitive to timing jitter**

M-ary signaling:

- **M-ary signaling** is a digital modulation technique where **multiple bits are transmitted per symbol**.
- Instead of sending **binary (2-level) signals**, M-ary uses **M distinct symbols**, where $M=2^k$ (k=number of bits per symbol).
- **Examples**: QPSK (M=4), 16-QAM (M=16), 64-QAM (M=64).

Key Concepts

Symbol Rate vs. Bit Rate:

- **Bit rate (R_b)**: Number of bits transmitted per second (bps).
- **Symbol rate (R_s)**: Number of symbols transmitted per second (Baud).
- Relationship: $R_b=R_s \times \log_2 M$.

Bandwidth Efficiency:

- Higher M allows more bits per symbol, improving spectral efficiency.
- However, higher M requires **higher SNR** (more susceptible to noise).

Types of M-ary Signaling

M-ary Amplitude Shift Keying (M-ASK)

- **Modulation**: Varies **amplitude** of the carrier.
- **Example**: 4-ASK uses 4 amplitude levels (00, 01, 10, 11).
- **Disadvantage**: Susceptible to amplitude noise.

M-ary Phase Shift Keying (M-PSK)

- **Modulation**: Varies **phase** of the carrier.
- **Examples**:
  - **QPSK (M=4)**: 4 phases (0°, 90°, 180°, 270°).
  - **8-PSK (M=8)**: 8 phases (45° apart).
- **Advantage**: More robust to noise than M-ASK.
- **Disadvantage**: Higher *M* requires precise phase detection.

M-ary Quadrature Amplitude Modulation (M-QAM)

- **Modulation**: Combines **amplitude and phase** variations.
- **Examples**:
  - **16-QAM (M=16)**: 4 amplitude levels × 4 phases.
  - **64-QAM (M=64)**: Used in modern Wi-Fi and 4G/5G.
- **Advantage**: Highest spectral efficiency.
- **Disadvantage**: Requires high SNR.

M-ary Frequency Shift Keying (M-FSK)

- **Modulation**: Uses **M different frequencies**.
- **Example**: 4-FSK (4 frequencies for 2 bits/symbol).
- **Advantage**: Robust in noisy channels.
- **Disadvantage**: Requires more bandwidth.

Performance Comparison

Modulation	**Bits/Symbol ($\log_2 M$)**	**Bandwidth Efficiency**	**SNR Requirement**
BPSK (M=2)	1	Low	Low
QPSK (M=4)	2	Medium	Medium
16-QAM (M=16)	4	High	High
64-QAM (M=64)	6	Very High	Very High

Advantages & Disadvantages

Advantages	**Disadvantages**
Higher data rate (more bits/symbol)	Requires higher SNR
Better bandwidth efficiency	More complex detection
Reduced symbol rate (lower bandwidth)	Increased BER at high *M*

Applications

- **Wireless Communication** (Wi-Fi, LTE, 5G use QAM).
- **Optical Communication** (QAM in fiber optics).
- **Digital TV Broadcasting** (QPSK, 16-QAM).

M-ary comparison with binary signaling:

Fundamental Differences

Feature	Binary Signaling	M-ary Signaling
Symbols	2 levels (0, 1)	M levels (M = 2^k)
Bits per Symbol	1 bit/symbol	k=log2*M* bits/symbol
Example Schemes	BPSK, OOK, 2-FSK	QPSK (M=4), 16-QAM, 64-QAM

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2. Bandwidth Efficiency

- **Binary Signaling**:
  - Transmits **1 bit per symbol** → Lower spectral efficiency.
  - Bandwidth requirement: $B \approx Rb$ (bit rate).
- **M-ary Signaling**:
  - Transmits **llog2*M* bits per symbol** → Higher spectral efficiency.
  - Bandwidth requirement: $B \approx \frac{R_b}{log_2 M}$

Example:

- For Rb=10 Mbps:
  - Binary (BPSK): $B \approx 10$ MHz.
  - 16-QAM ($\log_2 16=4$): $B \approx 2.5$ MHz.

3. Power Efficiency & SNR Requirements

Modulation	SNR per Bit (for same BER)
BPSK (Binary)	Lowest (most power-efficient)
QPSK (M=4)	Similar to BPSK (phase-based)
16-QAM (M=16)	~4× higher SNR than BPSK
64-QAM (M=64)	~10× higher SNR than BPSK

M-ary improves **bandwidth efficiency** but requires **higher transmit power** to maintain the same BER.

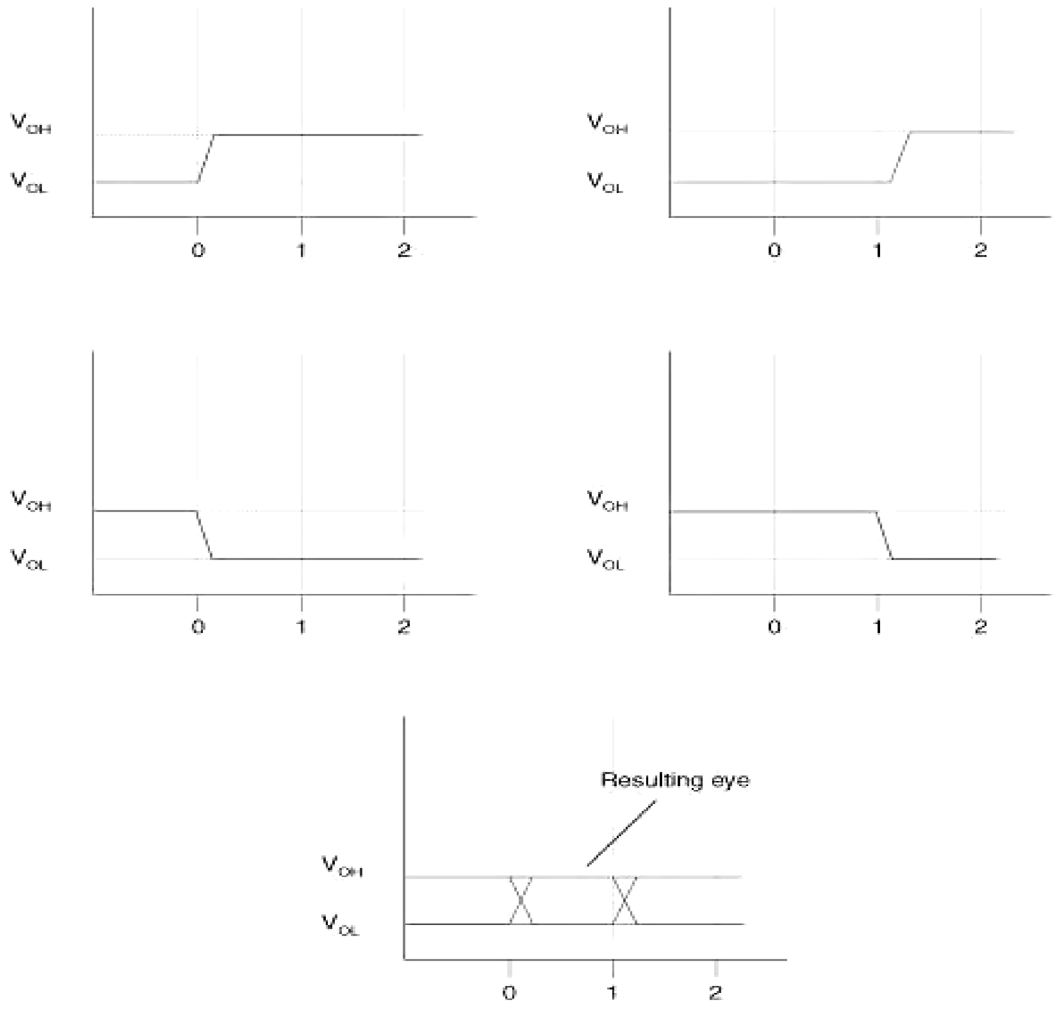
- **Choose Binary Signaling** if:
  - Power efficiency is critical (e.g., satellite links).
  - Channel conditions are poor (low SNR).
  - Simplicity is prioritized (e.g., IoT sensors).
- **Choose M-ary Signaling** if:
  - Bandwidth is limited (e.g., cellular networks).
  - High data rates are needed (e.g., 5G, Wi-Fi 6).
  - SNR is sufficiently high (e.g., short-range communications)

Eye Diagram

An **eye diagram** is a graphical representation of a digital signal's quality, created by overlaying multiple symbol periods. It resembles an "eye" due to the opening between signal transitions, providing key insights into:

- **Timing jitter**
- **Noise margins**
- **Intersymbol interference (ISI)**
- **Signal-to-noise ratio (SNR)**

➢ A properly constructed eye should contain every possible bit sequence from simple alternate 1's and 0's to isolated 1's after long runs of 0's, and all other patterns that may show up weaknesses in the design. Eye diagrams usually include voltage and time samples of the data acquired at some sample rate below the data rate. In **Figure 1** , the bit sequences 011, 001, 100, and 110 are superimposed over one another to obtain the final eye diagram.



- A perfect eye diagram contains an immense amount of parametric information about a signal, like the effects deriving from physics, irrespective of how infrequently these effects occur. If a logic 1 is so distorted that the receiver at the far end can misjudge it for logic 0, you will easily discern this from an eye diagram. What you will not be able to detect, however, are logic or protocol problems, such as when a system is supposed to transmit a logic 0 but sends a logic 1, or when the logic is in conflict with a protocol

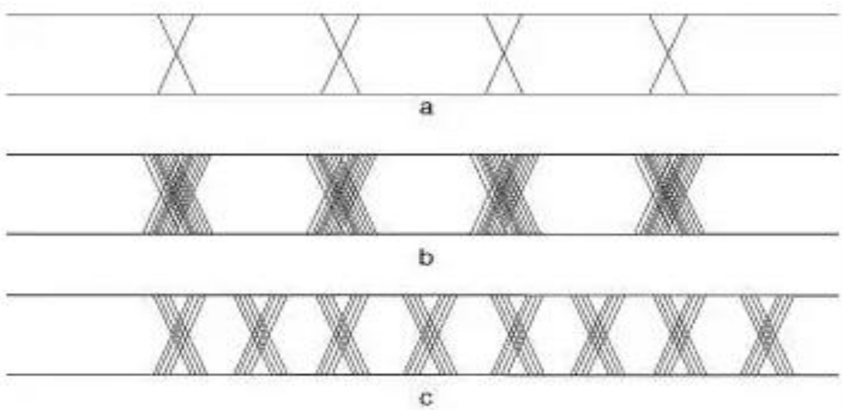


Figure 2 (a) Finite rise and fall times cause eye diagrams to look like this image rather than like a rectangle. (b) Jitter results from the misalignment of rise and fall times. (c) Although the absolute timing error or jitter margin is less than that in image b, this eye opening is smaller because of a higher bit rate.

Analyzing an eye diagram
As can be seen in **Figure,** an eye diagram can reveal important information. It can indicate the best point for sampling, divulge the SNR (signal-to-noise ratio) at the sampling point, and indicate the amount of jitter and distortion. Additionally, it can show the time variation at zero crossing, which is a measure of jitter.

Termination effect

The effect of termination is clearly visible in the eye diagrams generated. With improper termination, the eye looks constrained or stressed (**Figure 3a**), and with improved termination schemes, the eye becomes more relaxed (**Figure 3b**). A poorly terminated signal line suffers from multiple reflections. The reflected waves are of significant amplitude, which may severely constrict the eye. Typically, this is the worst-case operating condition for the receiver, and if the receiver can operate error-free in the presence of such interference, then it meets specifications.

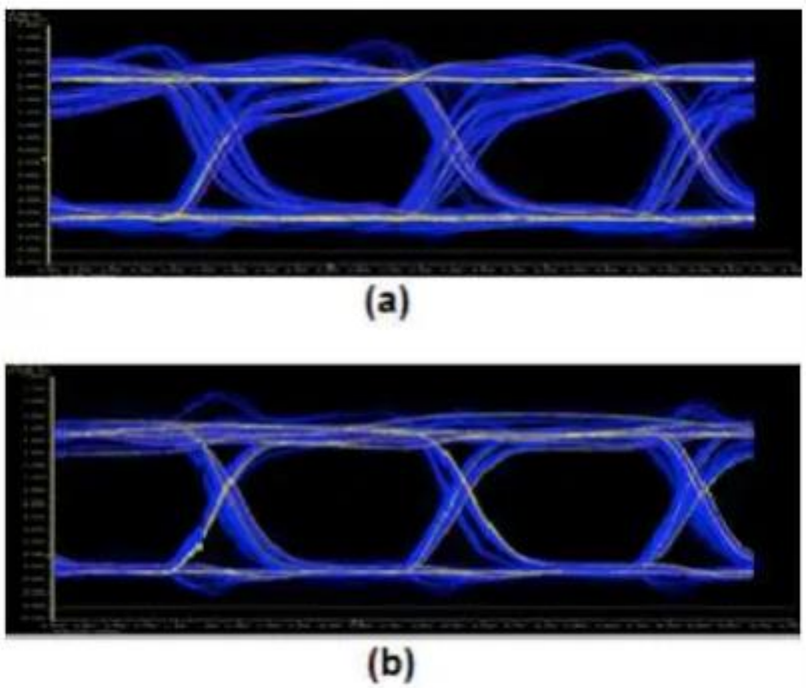


Figure 3 (a) Improper termination causes an eye diagram to look stressed. (b) Proper termination relaxes the eye.

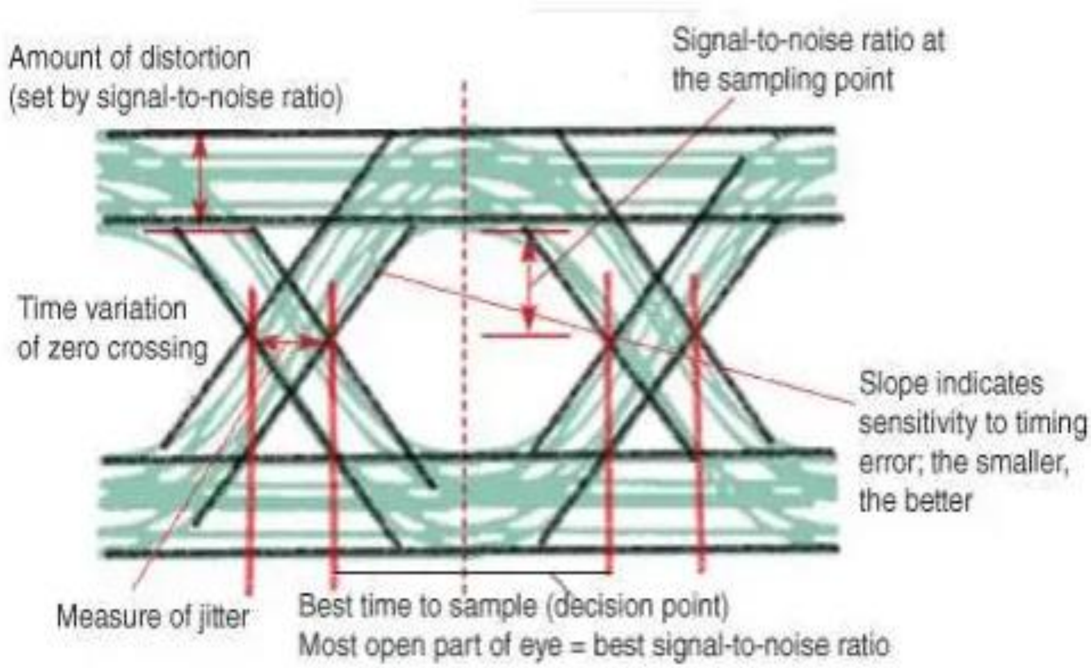


Figure An eye diagram can help you interpret a signal and determine the best time for taking a measurement.

- Eye diagrams provide instant visual data that engineers can use to check the signal integrity of a design and uncover problems early in the design process. Used in conjunction with other measurements such as bit-error rate, an eye diagram can help a designer predict performance and identify possible sources of problems.

The information provided by the eye patteren is

- The width of the eye opening defines the time interval over which the received wave can be sampled without error from intersymbol interference. It is apparent that the preferred time for sampling is the instant of time at which the eye is open widest.
- The sensitivity of system to timing error is determined by the rate of closure of the eye as the sampling time is varied.
- The height of the eye opening, at a specified sampling time, defines the margin over noise.